MIL-A-52679A
25 October 1983
SUPERSEDING
MIL-A-52679
11 March 1970

MILITARY SPECIFICATION

ANCHOR, SURFACING MEMBRANE: 12 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a steel picket type anchor to be used with membrane surfacing material for military airfields.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5680

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
TT-P-38	- Paint, Aluminum, Ready-Mixed.
PPP-B-601	- Boxes, Wood, Cleated Plywood.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-3627	- Penetrometers, Soil, Dial Indicating (Two Cones) and Spring Indicating (Airfield Cone).
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

DOD-STD-100	- Engineering Drawing Practices.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y14.5	- Dimensioning and Tolerancing.
ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualification.
ANSI/AWS A2.4	- Symbols for Welding and Nondestructive Testing, Including Brazing.
ANSI/AWS A5.1	- Covered Carbon Steel Arc Welding Electrodes.
ANSI/AWS A5.18	- Carbon Steel Filler Metals for Gas Shielded Arc Welding.
ANSI/AWS D1.1	- Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 109	- Steel, Carbon, Cold Rolled Strip.
ASTM A 615	- Deformed and Plain-Rolled-Steel Bars for Concrete Reinforcement.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The anchor shall be as shown in figure 1 and as specified herein. An anchor set shall consist of 26 anchors.

3.1.1 Interchangeability. All items shall be functionally and dimensionally interchangeable. The identification of part number and changes thereto shall be in accordance with the item identification and part number requirements of DOD-STD-100.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3.)

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3.3 Initial production. When specified (see 6.2), the contractor shall furnish the Government one or more anchor sets for inspection as specified in 4.4.

3.4 Materials. Materials shall be as specified herein and as shown on figure 1. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.7).

3.4.1 Material deterioration and control. The anchor shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4.2 Disc. The disc shall be steel conforming to ASTM A 109, temper No. 2, edge condition and finish optional.

3.4.3 Bar. The bar shall be steel conforming to ASTM A 615, grade 40.

3.4.4 Welding electrodes. Welding electrodes shall conform to ANSI/AWS A5.1, class E, 7015, 7016, or 7018 as required for type of welding performed. Metal inert gas filled core shall conform to ANSI/AWS A5.18, class ER70S-G.

3.5 Identification marking. The anchors shall be identified in accordance with MIL-STD-130 and shall be metal die stamped as shown in figure 1.

3.6 Treatment and painting. The anchors shall be cleaned, treated, and painted in accordance with MIL-T-704, type C. Paint shall conform to TT-P-38.

3.7 Workmanship. All parts of the anchor including stampings, machined surfaces, and welded parts shall be clean and free from dirt, fins, pits, scale,

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flux, and other harmful, extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.7.1 Welders and welding.

3.7.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ASME or AWS codes.

3.7.1.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Weld spatter and slag shall be removed prior to treatment and painting.

3.7.1.2.1 Weld criteria. Welding shall be performed as follows.

3.7.1.2.1.1 Cracks and other defects. There shall be no cracks. Undercut shall not exceed 5 percent of material thickness. Overlap shall not exceed 1/4 inch in any 3/4 inch of weld length.

3.7.1.2.1.2 Weld size. Fillet welds shall not exceed the next 1/16 larger size. Fillet welds shall not be undersize.

3.7.1.2.1.3 Weld contour. Flush contour of plug welds may be obtained by mechanical means.

3.7.2 Steel fabrication. Steel used in the fabrication of the anchor shall provide original quality surface finish and shall be free from kinks and sharp

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bends. Steel having an eroded surface is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Burned surfaces of flame-cut material shall be free of slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that the materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Initial production inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

d. Inspection comparison (see 4.7).

e. Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 Examination. The first article anchor set shall be examined as specified in 4.6.1. Presence of one or more defects on any of the anchors shall be cause for rejection.

4.3.2 Tests. The first article anchor set shall be tested as specified in 4.6.2.1. Failure of the test shall be cause for rejection.

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4.4 Initial production inspection. When specified (see 3.3), one or more initial production anchor set will be selected at random by the Government from the first 50 anchor sets being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production anchor set shall not exclude the remaining anchor sets from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production anchor set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production anchor set and shall be cause for refusal by the Government to continue acceptance of production anchors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on anchors previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all anchors accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level II for examination and level S-4 for tests. The unit of product shall be one anchor.

4.5.2 Examination.

4.5.2.1 Samples. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1. AQL shall be 4.0 percent defective for major defects. Presence of a critical defect shall be cause for rejection.

4.5.3 Tests.

4.5.3.1 Samples. Samples selected in accordance with 4.5.1 shall be tested as specified in 4.6.2. AQL shall be 4.0 percent defective.

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4.6 Inspection procedure.

4.6.1 Examination. The anchors shall be examined as specified herein for the following defects:

Critical

1. Edges sharp.

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Dimensions not as specified.
106. Identification markings not as specified or illegible.
107. Treatment and painting not as specified.
108. Workmanship not as specified.
109. Weld not as specified.

4.6.2 Tests.

4.6.2.1 Dynamic. Emplace each anchor in a soil having an average airfield index of 12-14 as determined by a Government-loaned penetrometer conforming to MIL-P-3627, type II, (see 6.5 and 6.6). The anchor shall be driven into the ground until bottom of the head is flush with the ground. Remove and examine. Weld failure or evidence of weld cracking shall constitute failure of this test.

4.7 Inspection comparison. The Government may select anchor sets at any time during the contract production period and subject these anchor sets to the examination specified in 4.6.1 and the tests specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on anchor sets selected at random from those which have been accepted by the Government and will not include the previously inspected first article and initial production anchor sets. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison anchor set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison anchor sets and shall be cause for refusal by the Government to continue acceptance of production anchor sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on anchor sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all anchor sets accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

110.	Container not as specified for level A.
111.	Arrangement in container not as specified for level A.
112.	Strapping not as specified for level A.
113.	Packing not in accordance with the referenced document as specified for commercial.
114.	Marking missing, illegible, incorrect, or incomplete for level A or commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The anchors shall be arranged and packed in accordance with figure 2.

5.1.2 Commercial. The anchors shall be packed in accordance with MIL-STD-1188.

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5.2 Marking (see 6.4).

5.2.1 Military. Marking for level A packaging shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The anchors are intended to be used to hold down airfield membrane surfacing material for tactical military airfields.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval, and the number of units required (see 3.2).

c. When initial production inspection is required and number of anchor sets to be furnished, when applicable (see 3.3).

d. Degree of packing required (see 5.1).

6.3 First article. When a first article inspection is required the item shall be examined and should be a first produced anchor. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and approval of the document's first article.

6.4 Packaging. Degrees of preservation have not been included as the anchors do not require additional preservation. For purposes of preservation/packing level marking the preservation shall be designated as level A.

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 4.6.2.1.

6.6 Determination of airfield index.

6.6.1 Description, use and maintenance of instrument.

6.6.1.1 Description. The airfield cone penetrometer is a probe-type instrument designed to measure soil strength. It consists of a right circular cone with a base diameter of 1/2 inch mounted on a graduated staff; on the opposite

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end of the staff are a spring, a load indicator, and a handle. The overall length of the assembled penetrometer is about 36-1/8 inches. For ease in carrying, the penetrometer can be disassembled into three main pieces: Two extension staffs, each 12-5/8 inches long, and one piece 14-3/4 inches long containing the cone, handle, spring, and load indicator.

6.6.1.2 Use. Before the penetrometer is used, the instrument should be inspected to see that all joints are tight and that the load indicator reads 0. To operate the penetrometer, the hands are placed symmetrically on the handle, palms down. Force is applied to the handle (with the operator's arms steadied against his thighs) until a slow, steady, downward movement of the instrument occurs. The load indicator is read at the moment the base of the cone enters the ground (surface reading) and at desired depths at the moment the corresponding depth mark on the shaft reaches the soil surface. The reading is made by shifting the line of vision from the soil surface to the indicator just a moment before the desired depth is reached. Maximum efficiency is obtained with a two-man team in which one man operates and reads the instrument while the other acts as recorder. It is possible for one man to operate the instrument and record the measurements by stopping the penetration at any intermediate depth, recording previous readings, and then resuming penetration. Observance of the following rules is important in obtaining accurate data:

a. The instrument should read 0 when suspended by the handle and 15 when a 150-pound load is applied.
b. The instrument should be kept in a vertical position while it is in use.
c. The rate of penetration should be about 1/2- to 1-inch per second; however, slightly faster or slower rates will not materially affect the readings.
d. If it is suspected that the cone is encountering a stone or other foreign body at the depth where a reading is desired, another penetration should be made nearby.
e. The readings should be taken at the proper depths. Carelessness in determining depth is one significant source of error in the use of the penetrometer.

6.6.1.3 Maintenance. The airfield cone penetrometer is simply constructed of durable metals and needs little care other than cleaning and oiling. The calibration should be checked occasionally. As noted earlier, the load indicator should read 0 when the instrument is suspended by the handle and 15 when a 150-pound load is placed on the handle. If an error in excess of about 5 percent is noted, the penetrometer should be recalibrated.

6.7 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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Custodians:
Army - ME
Navy - YD
Air Force - 99

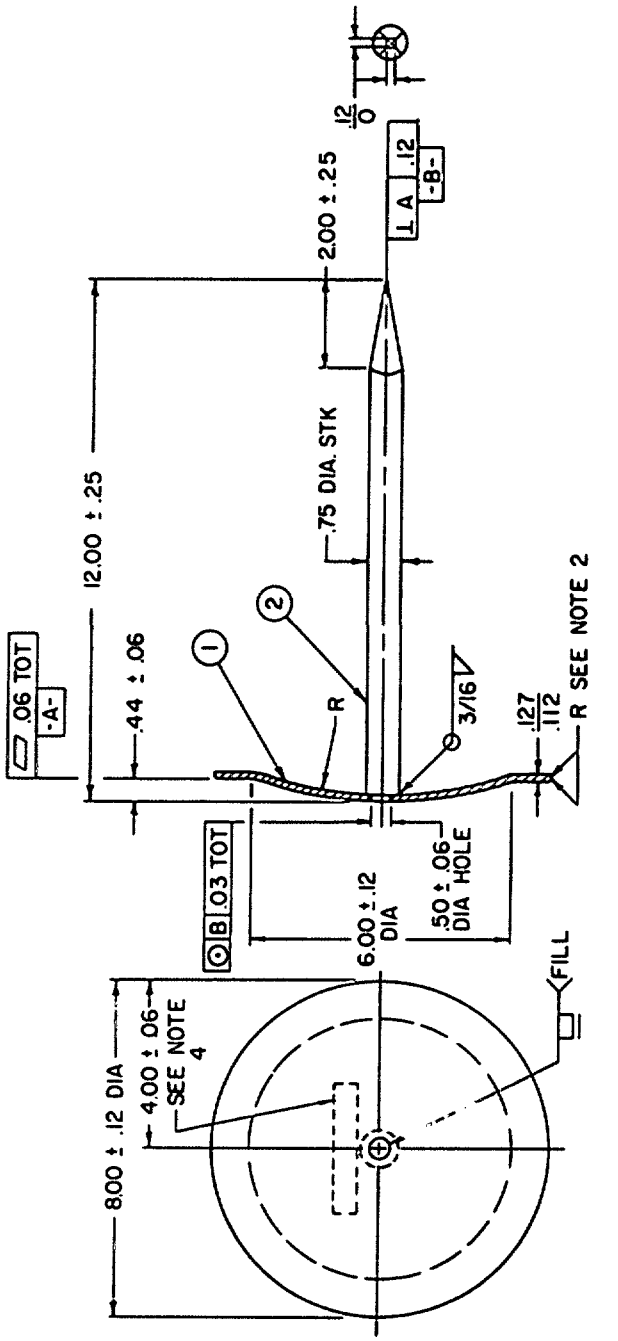
Review activity:
Air Force - 84

User activity:
Navy - MC

Preparing activity:
Army - ME

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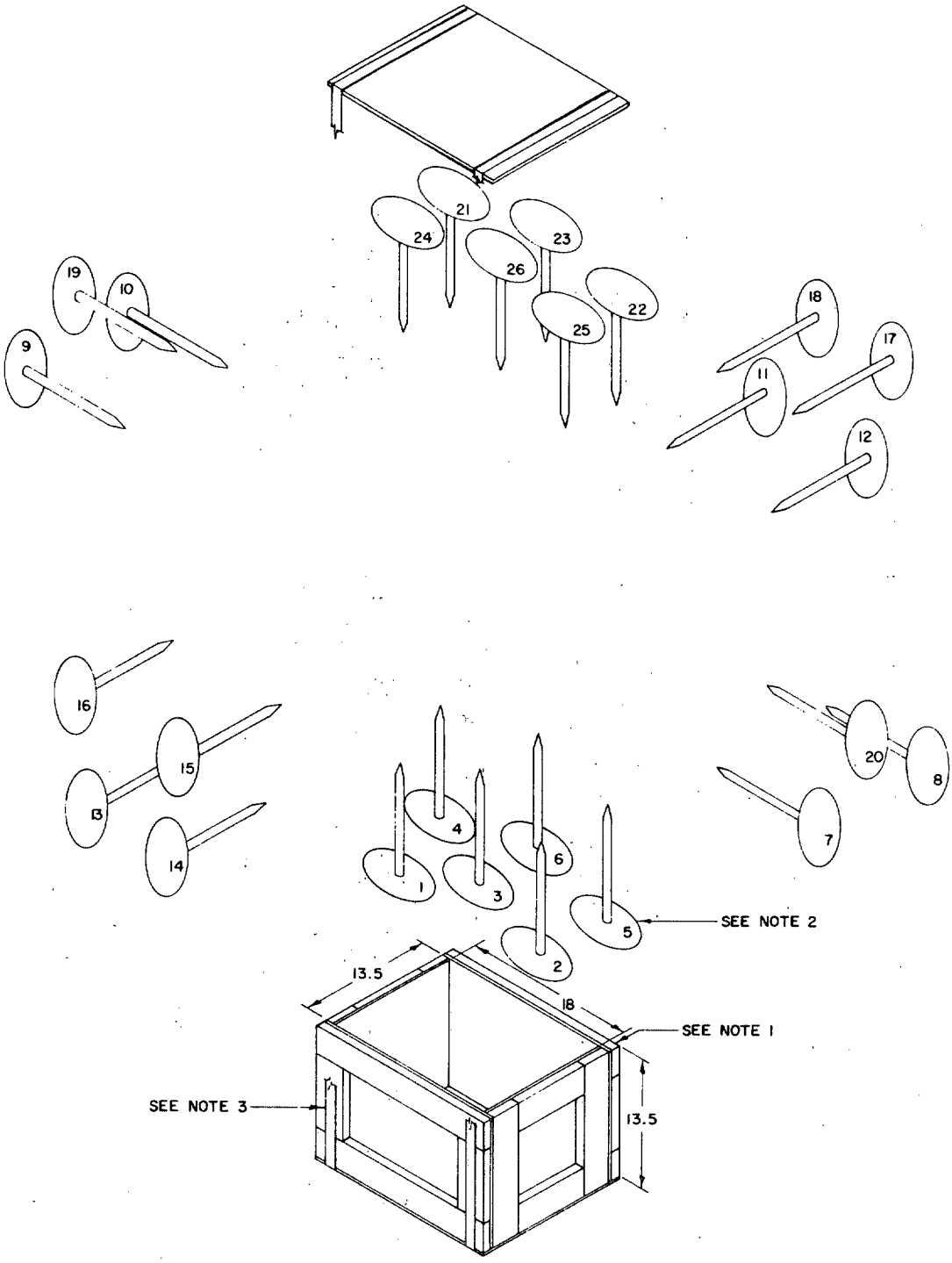


FIND NO	SYM	CODE IDENT	DWG SIZE	PART OR IDENTIFYING NO.	QTY REQD	NOMENCLATURE OR DESCRIPTION	SPECIFICATION	MATERIAL
1				TEMP 2 EDGE	1	DISC .125 STK NOM.	ANSI/ASTM A109	STL
				AND FINISH OPTL				
2				GR 40	1	BAR, REINFORCING NO. 6.	ANSI/ASTM A615	STL
3	B				AR	ELECTRODE, WELDING	SEE NOTE 3	STL

NOTES:
1. TREATMENT AND PAINTING, PER MIL-T-704 TYPE C. FINISH COAT PAINT PER TT-P-38.
2. GRIND ROUND AND DEBURR.
3. WELDING ELECTRODE.ANSI/AWS A5.1 CL E7015, E7016, E7018 OR ANSI/AWS A5.18 CL ER70S-G.
4. METAL DIE STAMP LETTERS .25 HIGH. (A) FEDERAL STOCK NO. (B) NOMENCLATURE (C) CONTRACT NUMBER (D) MFGR NAME (E) U.S.
5. FOR INTERPRETATION OF: WELD SYMBOLS SEE ANSI/AWS A2.4 DIMENSIONS AND TOL. SEE ANSI Y14.5
6. DIMENSIONS ARE IN INCHES.

FIGURE 1. Anchor, surfacing membrane, 12 inch. X1902A

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NOTES:
1. BOX IN ACCORDANCE WITH PPP-B-601, OVERSEAS TYPE, STYLE I.
2. ANCHORS MUST BE ARRANGED IN BOX BY SEQUENCE NUMBER 1 THRU 26.
3. STRAPPING IN ACCORDANCE WITH QQ-S-781, CLASS 1, TYPE I OR IV, FINISH B, SIZE .035 x 3/4.
4. DIMENSIONS ARE IN INCHES.

FIGURE 2. Packing arrangement and detail.

X-1903A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-52679A	Anchor, Surfacing Membrane: 12 Inch

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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